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Palaeomagnetism of Palaeozoic rocks from the Cabo de Penas, Asturia, Spain

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Summary. Ordovician volcanic samples of the Cabo de Peñas area yielded a primary remanent magnetization ($D = 202°$, $I = 78°$, $\alpha_{95} = 6°$) which is partially or entirely overprinted by a Late Palaeozoic remagnetization. Late Devonian–Early Carboniferous sedimentary samples reveal a pre- or syntectonic Carboniferous remanent magnetization similar to the secondary one of the volcanic samples. A soft post-tectonic component appears also in few samples, for which a Late Carboniferous age is inferred. Comparison of the Carboniferous remanent magnetizations and structural parameters suggests that in a late stage of the deformation, the sites have undergone minor rotations around a vertical axis. A 'strike correction' is therefore proposed to take such an effect into account. The average of the corrected mean-site directions ($D = 178°$, $I = 19°$, $\alpha_{95} = 9°$) is considered to be a good approximation of the Early or Middle Carboniferous palaeomagnetic direction in Asturia.

Introduction

Since the concepts of plate tectonics have been introduced, many authors from different fields of the Earth sciences have shown an interest in the geodynamic evolution of the Atlantic ocean and its bordering continents. Among the various methods used, one of the most powerful is palaeomagnetism. The demonstration of the opening of the Atlantic ocean using palaeomagnetic data (Runcorn 1956) is a very good example of the utility of such analyses. While the Mesozoic and Cenozoic history of these areas has been well characterized a great deal of work is still needed to define the relevant Palaeozoic plate dynamics. Palaeomagnetic studies from cratonic North America, Africa and Europe have permitted the construction of a general outline of the movements of the major plates involved (Van der Voo 1979, 1982; Kent 1980). In the reconstructions, a new concept was introduced: the 'Amorica plate' (Van der Voo 1979). This microplate includes at least the Armorican Massif and may be extended to the Bohemian Massif, English Cornwall, the Avallon zone in North America, the Iberian Meseta and perhaps North Morocco. Armorica may have played an essential role in the Hercynian orogeny, because of its situation between the three major

plates, Laurentia, Baltica and Gondwana. In order to investigate the interaction between these plates, and the 'border effects', we have to study areas within the orogenic belts themselves. A first review of such work has been made recently (Van der Voo & Channel 1980), showing the strength of the method. However, great care is required for the interpretation of the data, and limits are imposed by problems of overprinting, deformation or metamorphism (Perroud 1980). The Iberian Meseta is such an area, in the heart of the large Variscan (= Hercynian) system of Western Europe. Although there is an abundance of suitable material in this key area, so far few palaeomagnetic data have been obtained.

A major structural feature, the Ibero-Armorican arc (IAA) links the Armorican and Iberian zones across the Bay of Biscay. This arc is defined by structural lines, palaeontological units and petrological zonation (Bard, Capdevila & Matte 1971; Cogné 1971; Matte 1974). Many geologists consider this arc as a critical area in the discussion of the numerous models proposed for the Hercynian orogeny. In essence either a primary origin (before sedimentation of Palaeozoic beds) or a secondary (tectonic) origin of the arc could be envisioned. Stratigraphical and palaeontological arguments (Henry *et al.* 1974; Paris & Robardet 1977) as well as structural studies (Julivert 1971) and even the first palaeomagnetic directions obtained in the Asturian part of the arc (Ries, Richardson & Shackleton 1980; Bonhommet *et al.* 1981), indicate a partially secondary origin. Better determined Carboniferous or older directions are needed to quantify more precisely this phenomenon. In this context we will present here a detailed palaeomagnetic study made in the Cabo de Peñas area (northern Asturia, Spain). Because the results obtained in this work have already been used in a synthetic interpretation with other data from the IAA (Perroud & Bonhommet 1981) we will emphasize here the palaeomagnetic analysis rather than the tectonic interpretation already given in that previous publication.

Geological setting and sampling

The Cabo de Peñas region lies on the North Coast of Spain where the northern limb of the Asturian arc is truncated by the sea. In the cliffs from Cabo de Peñas towards the southeast (Fig. 1) an almost complete succession exists from Lower Ordovician till Lower Carboniferous (Truyols & Julivert 1976). The 'Barrios' quartzite is overlain by Ordovician shales, with an intercalation of volcanic material in the lower part. Above, the Ordovician series end up with a volcano-detritic succession, including basaltic lavas in the upper part. Then, there is an uninterrupted period of sedimentation up to Early Carboniferous times, with large amounts of fossils giving well-determined stratigraphical ages. Several periods of red sedimentation occur in the sequence. In particular the Siluro-Devonian 'Furada' sandstone (called San Pedro elsewhere) is associated with iron ores and has not been sampled because it is unlikely that these complex ores will yield usful palaeomagnetic information. The whole area has been folded by Hercynian deformation and has a general NE–SW trending synclinal shape. Locally, the strata are strongly deformed (Julivert 1976), precluding any sampling. The minor fold axes are always subhorizontal, but their directions are not constant. The particular tectonic context of the Asturian arc (Julivert 1971) with its second phase of folding radial to the arc, which is associated with the tectonic closure of the arc, may be the cause of this dispersion.

Eight sampling sites were chosen in the most suitable zones for our purpose and are distributed in the following manner: Ordovician volcanism – two sites (I in the first volcanic intercalation and J in the basaltic flows), 31 cores; Late Devonian Candas red sandstones – three sites (C, D and F), 32 cores; Early Carboniferous Alba redgrits – three sites (A, B and E), 23 cores. As shown in Fig. 1, sampling of the sediments has been carried out on both

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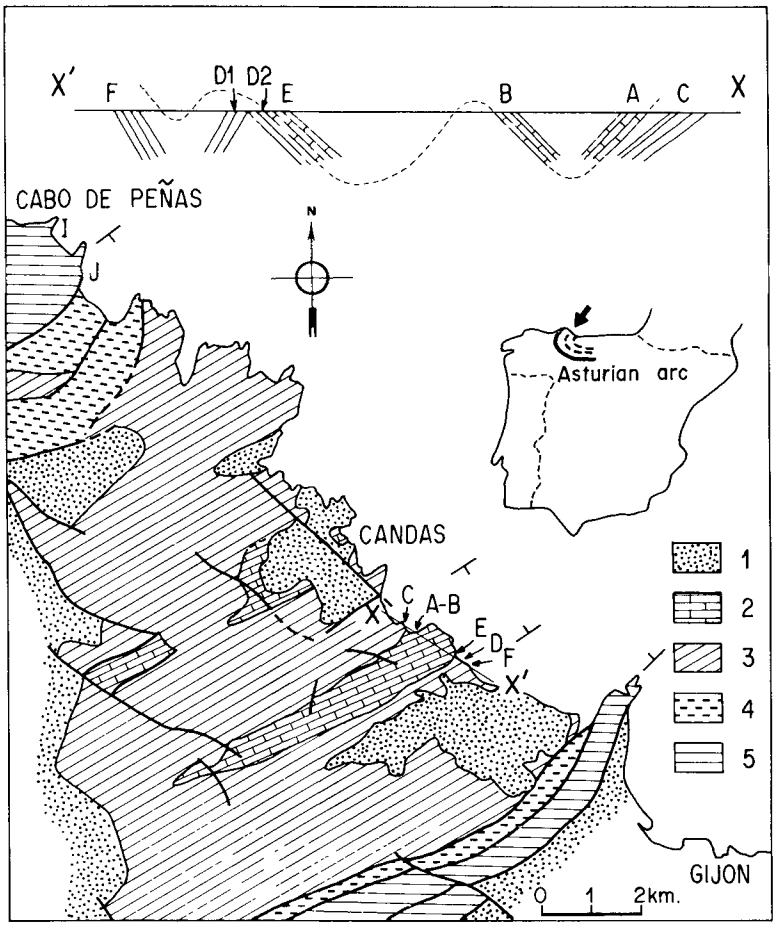


Figure 1. Schematic geological map of the Cabo de Peñas area, with the sampling sites (A–F and I, J); X–X′ is a cross-section showing the structural position of sites A–F on both limbs of folds. 1 is for the Mesozoic cover, 2 for Carboniferous, 3 for Devonian, 4 for Silurian and 5 for Ordovician.

limbs of folds roughly parallel to the main structural trend. One site (D) is composed of 2 groups (D1 and D2) of seven cores each from the two limbs of a 10 m wide minor fold. In two other sites (B and E), the structural position varies slightly within the site.

Microscope studies of all the representative rock types do not reveal any traces of metamorphism or of neoformed opaque minerals. Finally, we should point out that the two sampled sedimentary formations normally succeed one another, which means that their stratigraphical ages are very close; we will therefore interpret their directions together.

Laboratory techniques and measurements

The magnetic measurements have been done either with a Schonstedt computer-assisted spinner magnetometer (University of Rennes) or with an ScT cryogenic magnetometer (University of Michigan). Demagnetization procedures have been carrried out with Schonstedt equipment [alternating field (AF) demagnetizing apparatus GSD-1 and non-magnetic furnace TSD-1, both in mumetal shields with a very low residual field of about 10 nT]. Progressive Isothermal Remanent Magnetization (IRM) acquision experiments

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have been performed with a Brucker BE-10 electromagnet. A Digico magnetometer adapted for temperature cycles between −196° and 700°C has been used to obtain the intensity versus temperature behaviour of a laboratory-created IRM. Before any treatment, cores were sawed into standard specimens, and a viscosity test (Thellier & Thellier 1959) has been performed for all the specimens.

Palaeomagnetic results

ORDOVICIAN VOLCANISM

The intensity of the NRM is high (0.5 A m^{-1} for site J) and uniform within each site. The 8 day viscosity coefficient [intensity of Viscous Remanent Magnetization (VRM) produced during 8 days in the local field divided by the Natural Remanence (NRM)] is very low (5 per cent for site J, 10 per cent for site I). Directions are fairly well clustered, downward

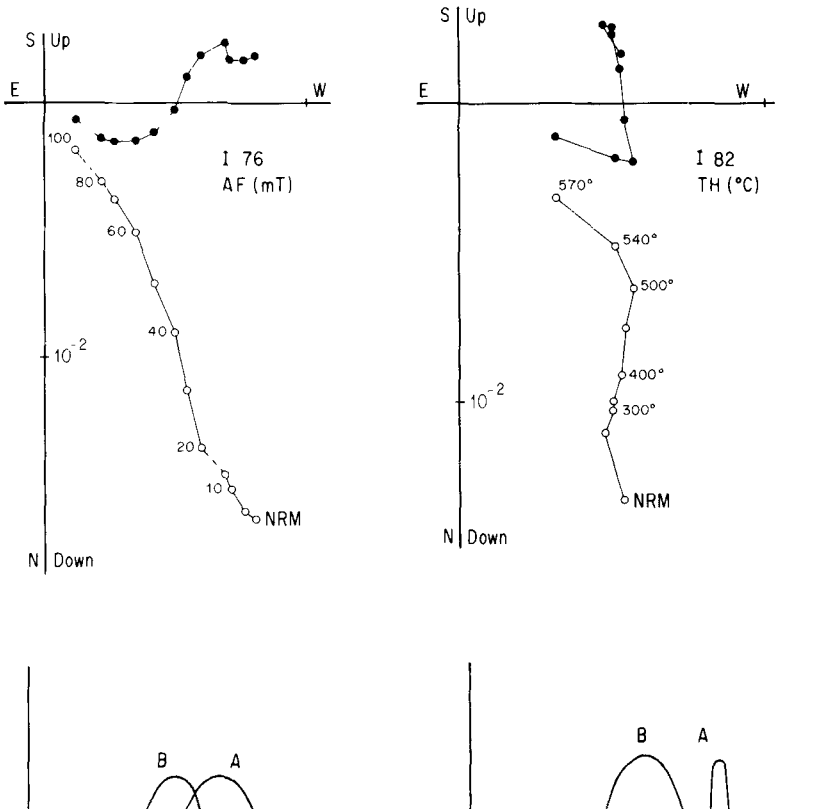


Figure 2. Zijderveld demagnetization diagrams for site I. Close (open) dots are for projection on the horizontal (vertical) plane. Units along axes are in A m^{-1}. Bottom: overlapping of both coercivity and temperature spectra estimated for components A and B. The arrows indicate the area of separation for each component.

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and to the south. Stepwise AF and thermal demagnetizations of specimens of site I clearly show a two-component behaviour after removal of a very soft component at 300°C or 20 mT (Fig. 2). The high-coercivity component corresponds to the high unblocking temperature one. The separation of the components seems to be successful with the two methods together. As a matter of fact, thermal treatment isolates very well the lower unblocking temperature components, leading to a low within-site dispersion ($k = 45$). On the other hand, AF does not isolate this component well, as shown in Fig. 2, but instead reveals easily the hard-coercivity component leading to very clustered directions for the whole site ($k = 50$). This behaviour may be explained in terms of overlapping of coercivity and thermal spectra as shown by Fig. 2. In the AF procedure the spectrum for component A is quite large compared to the B spectrum; the only range of univectorial decay is therefore for the high fields (indicated by the double arrow in Fig. 2). Inversely, in thermal treatment this range appears for the low temperatures. The behaviour of the samples for the whole site is very uniform, as shown on the intensity versus temperature diagram (Fig. 3b), which indicates unblocking temperatures typical for magnetite (or titano-magnetite with low Ti). IRM acquisition and low-temperature experiments confirm the presence of magnetite as the main if not unique carrier of the remanence for both sites I and J (Fig. 3a, c); saturation of the IRM is reached at 0.1 tesla for both sites I and J. On the other hand, reheating to room temperature of an IRM acquired at −196°C reveals a very nice (multidomain)

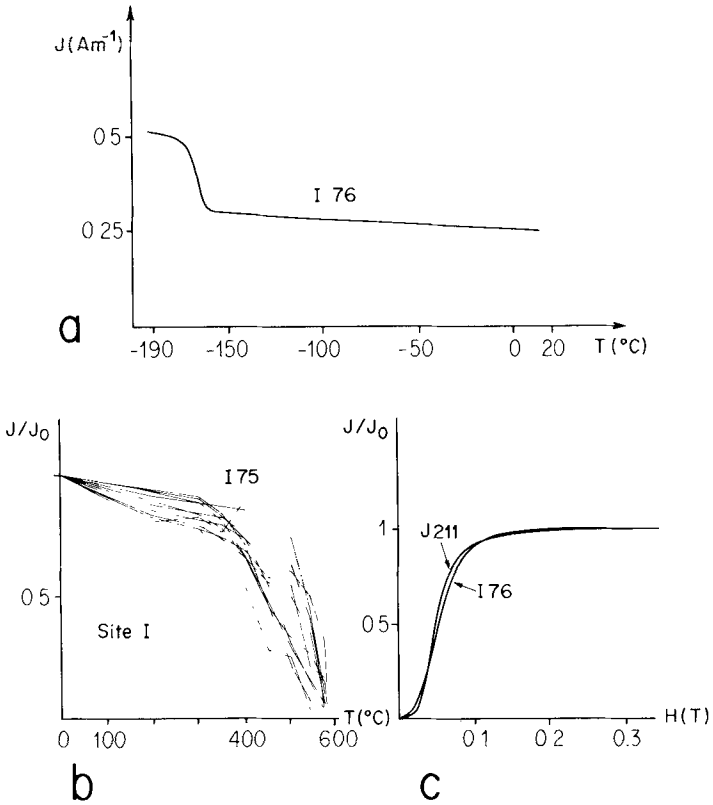


Figure 3. (a) Continuous measurement of intensity versus temperature during the reheating up to room temperature of an IRM acquired at −196°C. (b) Normalized intensity versus temperature curves during stepwise thermal demagnetization for all samples of site I. (c) IRM acquisition curve for samples of both site I and J in fields up to 0.3 T.

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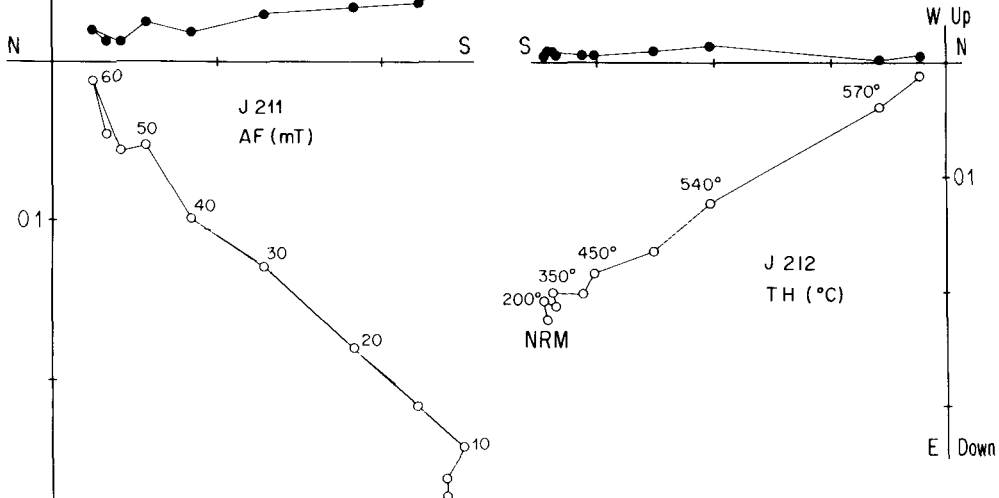


Figure 4. Zijderveld diagrams of stepwise AF and thermal demagnetization procedures for site J (same conventions as in Fig. 2).

magnetite transition around −160°C for a specimen of site I. Demagnetizations of specimens from site J show quite different behaviour: here, only one component is observed in both thermal and AF demagnetization which decreases systematically towards the origin in the Zijderveld (1976) diagrams (Fig. 4). In alternating fields above 60 mT peak value, the samples show a zigzag-shaped curve. We infer that this results from the introduction of anhysteretic components, and that the measurement procedure (the sample is inverted in the coil at each step) leads to this kind of curve; therefore AF demagnetization was discontinued after 60 mT. However, these samples have medium coercivities so stable directions may be determined both by AF and thermal demagnetizations; indeed, mean directions obtained by the two methods are identical in a statistical sense.

The three mean directions with their statistical parameters are given in Table 1, before and after tectonic correction, and shown in Fig. 5, with their circles of confidence. The three *in situ* directions are significantly different from the present-day field and from all

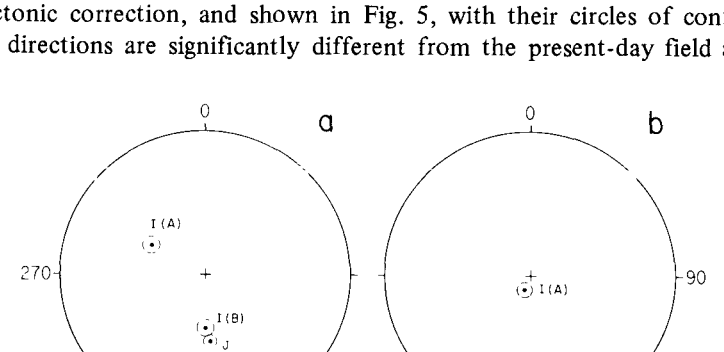


Figure 5. Stereographic projections of the mean directions for site I and J before (left) and after (right) tilt correction.

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Table 1. Site-mean directions after AF and thermal cleaning (S = soft component on sites B, C, D1, E, F).

Component type of rock		Interval AF (mT)	Interval Th(°C)	N	Before tilt correction Dec.	Inc.	k	α_{95}	After tilt correction Dec.	Inc.	k	α_{95}	Inferred age magnetization
I (A)	volc.	60–100	500–570	13	298	+45	50	6	202	+78	50	6	Ordovician
B	grit	30–100	300–450	6	176	+2	227	4	181	+27	274	4	
C	sandst.		500–680	9	170	+43	75	5	156	+18	75	5	
D1	sandst.		500–680	7	208	+42	114	5	186	+2	114	5	Early or
D2	sandst.		500–680	7	181	−15	290	3	183	+22	290	3	Middle
E	grit	30–100	300–500	7	182	−14	33	9	193	+37	154	4	Carboniferous
F	sandst.		500–680	7	177	−17	262	3	181	+26	262	3	
I (B)	volc.	20–60	350–500	11	179	+49	45	6	162	+9	45	6	
J	volc.	20–60	350–570	11	175	+40	104	4	160	+7	104	4	
S	grit + sandst.	10–60	0–500	8	185	+23	27	10	196	+21	4	–	Late Carboniferous

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post-Hercynian directions known for Europe (after closing of the Bay of Biscay, Le Pichon & Sibuet 1971). On the other hand, rock magnetism experiments reveal that a single magnetic phase may carry the different components. Moreover, optical examinations suggest that no recrystallization has taken place in these rocks. This evidence combined with the magnetic behaviour outlined above suggests that the I(B) component is a partial thermoremanent magnetization which has partially overprinted an older I(A) component. The mean direction for site J is similar to the one for I(B), especially after the tilt correction. Although it is impossible to make a statistically significant fold test with two entries, one could observe that the angular scatter between I(B) and J mean direction decrease from 9.5° to 3° when applying the correction. Their southerly and shallow direction is compatible with Late Palaeozoic data. The natural and simplest interpretation which is derived from this analysis is that I(A) is a primary Ordovician magnetization and that I(B) and J are partial to complete Late Palaeozoic overprints. These last components do not look like Mesozoic or Cenozoic magnetizations, and they may have been acquired either before or perhaps during the folding. The A component at site I is therefore interpreted as Ordovician and primary, with a southerly steeply down direction. Such high inclinations have also been found in other Early Palaeozoic studies from the Armorica plate (Duff 1979, 1980; Perroud & Bonhommet 1981; Perroud, Bonhommet & Robardet 1982).

LATE PALAEOZOIC SEDIMENTS

Specimens from sites A to F show a very uniform magnetization within each site. Short-period viscosity is low and directions are always different from the present-day field. IRM acquisition experiments on Alba redgrit samples (Fig. 6) in a cryogenic magnet up to 5 tesla (University of Michigan) reveal a typical haematitic composition of the magnetic phase. Subsequent heating to 120°C shows only a small decrease of the saturation IRM implying that the contribution of goethite to the IRM is low, in spite of the fact that goethite has been observed as reaching up to 5 per cent of the total composition of the rock (Bonhommet *et al.* 1981). Samples of the Candas red sandstones have also typical haematite coercivity characteristics.

Alternating field demagnetizations for the Alba samples only, and thermal treatment for both formations have been carried out and typical Zijderveld plots are given in Fig. 7. A soft component (S), destroyed in low peak field values or low temperatures is found in eight samples from five sites, whereupon the demagnetizations indicate univectorial decay towards the origin for all samples from all sites except site A. From the specimens of site A no

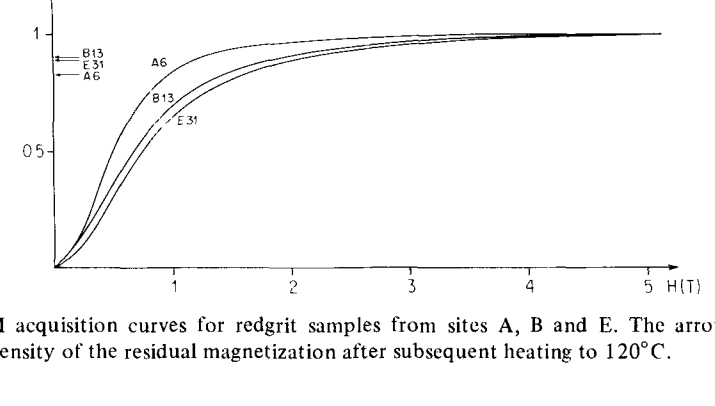


Figure 6. IRM acquisition curves for redgrit samples from sites A, B and E. The arrows indicate the normalized intensity of the residual magnetization after subsequent heating to 120°C.

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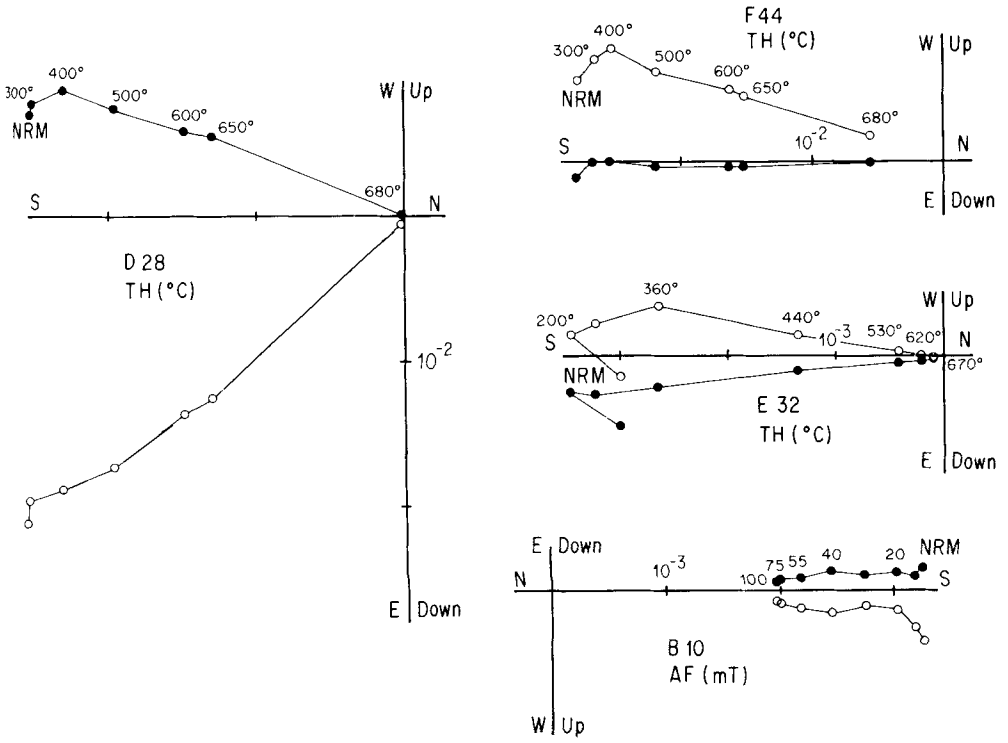


Figure 7. Zijderveld diagrams for Alba and Candas samples (same conventions as in Fig. 2).

characteristic direction could be obtained. The mean directions from the other six sites (B, C, D1, D2, E, F) and their circles of confidence are given before and after the tilt correction in Table 1 and are shown in Fig. 8(a, b). Individual directions of the soft component appear before and after tilt correction in Fig. 8(c, d). The mean direction is given in Table 1. This component clearly reveals a negative fold test at the 95 per cent confidence level (k decreases from 27 to 4 upon tilt correction). It is therefore interpreted as post-folding. Comparison with post-Hercynian European directions (after closing of the Bay of Biscay) shows that the most likely age for this component is Middle to Late Carboniferous. The directions of the hard component, however, satisfy several criteria for a pre-tectonic origin. Sites B ane E, for instance, for which the structural attitude varies within the site, show better clustering after the correction (Table 1). The angular scatter between the two mean directions of sites D1 and D2 decreases from 60° to 20° although they bypass one another. Finally, the Fisher (1953) statistics for the formation mean direction (six sites) show an increase of k from 7 (before) to 24 (after the tilt correction) giving a positive fold test at the 95 per cent confidence level. The age of the magnetization is therefore considered to be between Late Devonian to Early Carboniferous (age of the sedimentation) and Middle Carboniferous (age of the folding).

Discussion

As stated in the previous section, sites I and J present a secondary pre- to syntectonic magnetization. Its low inclination suggests a Late Palaeozoic age. For these reasons, we will now discuss the corresponding directions together with the magnetizations of the Late

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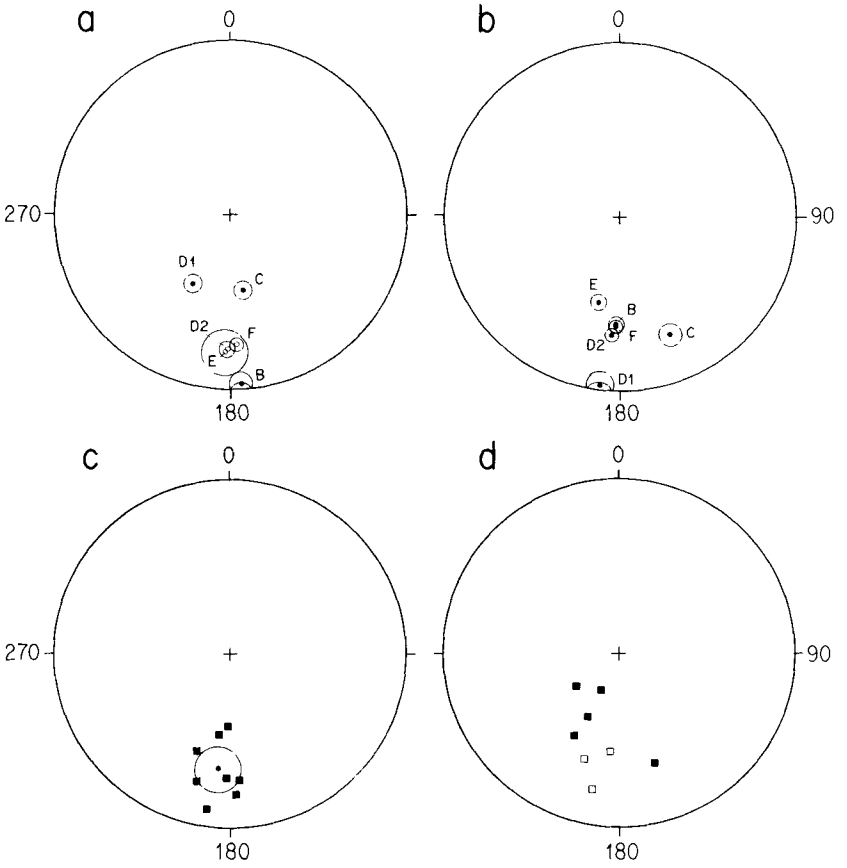


Figure 8. (a, b) Stereographic projections of the site-mean directions for Alba and Candas samples before (a) and after (b) tilt correction with their circles of confidence. (c, d) Individual data of the soft components before (c) and after (d) tilt correction. In (c) the mean direction is also indicated with its circle of confidence.

Palaeozoic sediments. Table 2 summarizes the eight Carboniferous site-mean directions with their radius of the circle of confidence (α_{95}), and the mean structural position of each site. It appears that the within-site scatter is always very low ($< 6°$) but that the between site dispersion is rather large by comparison. The declination varies from 156° to 193° and the inclination from 2° to 37°. [Note that the I(B) and J mean direction are also within these ranges.] This is not a satisfactory result, since the between site scatter should be of the same order of magnitude as the within-site one. In order to explain this situation, we must examine more deeply the distribution of the data and the parameters which might produce such dispersion. Among these are the structural parameters. Let us recall here that we may have some good reasons to expect independent dispersions for the declination and inclination of magnetic vectors in such an area where the folding has occurred at least in two non-parallel phases. For example, the tilt correction introduces a declination error when fold axes are plunging (MacDonald 1980), whereas inclinations are not disturbed. Inversely, internal deformation by shortening of the layers (Morash *et al.* 1981) may introduce inclination variations. It is therefore valuable to examine these parameters independently. Let us first look at the inclination dispersion.

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Table 2. Comparison of the pre- or syntectonic magnetization with the structural position, and 'strike' corrected directions for each site where a Carboniferous component has been revealed. (D, S) is for the declination-strike angle.

Site	Direction of magnetization: 100 per cent unfolded		Direction of magnetization: 70 per cent unfolded		Structural attitude Strike	Dip	(D, S) 100 per cent unfolded	(D, S) 70 per cent unfolded	After vertical axis rotations 100 per cent unfolded		After vertical axis rotations 70 per cent unfolded	
B	181	27	179	20	231	32NW	50	52	180	27	178	20
C	156	18	158	26	210	34SE	54	52	176	18	178	26
DI	186	2	188	15	235	61SE	49	47	181	2	183	15
D2	183	22	181	11	241	43NW	58	60	172	22	170	11
E	193	37	184	24	243	61NW	50	59	180	37	171	24
F	181	26	176	14	225	57NW	44	49	186	26	181	14
I (B)	162	9	164	22	224	48SE	62	60	168	9	170	22
J	160	7	162	18	212	45SE	52	50	178	7	180	18
Mean	175	19	174	19			52.4 ± 5.5	53.6 ± 5.3	178	19	176	19
k	22		48						39		126	

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We have seen before that these characteristic magnetizations present several criteria for a pre-tectonic origin. However, it can be argued that the inclination dispersion is anomalously large, and that directions of sites from NW dipping and from SE dipping limbs bypass each other during the tilt correction (see for example D1 and D2, Table 1). This situation could correspond to a magnetization acquired early after the beginning of the folding. As a matter of fact, a 70 per cent unfolding leads to a Fisher parameter (k) of 48 for the mean of the eight directions, instead of 22 for the complete (100 per cent) correction (70 per cent unfolding has been chosen since it actually maximizes the k value) (Table 2). In our particular case, this partial unfolding does not affect the declinations since those portions of the small circle around the rotation axis to which the site-mean directions are constrained have a near-constant declination. Therefore, while site-mean directions pass the fold test at the 95 per cent confidence level, careful inspection of the data allows up to propose two alternative interpretations: (1) either the magnetization has been acquired early during the folding (Middle Carboniferous) and the 70 per cent unfolding gives the best estimate of the dispersion or (2) the magnetization is truly pre-tectonic (Early to Middle Carboniferous) and the inclination dispersion has a different cause, such as internal deformation of the rock. It has been shown previously (Perroud 1982) from a comparison of the dip of the strata and the inclinations that all the sites with NW dip have an inclination greater than the mean (19°) and all the sites with SE dip have inclinations smaller than the mean. Moreover, the sites where the inclination shows greater deviations from the mean area those sites where the dip is also greater. As rigid folding rotations do not modify the inclination of pre-tectonic magnetizations with respect to the bedding, whereas structural position correlates well with the site-mean inclination, we tentatively suggested that internal deformation may have occurred in these beds which may have dispersed somewhat the magnetic directions. However, clear evidence for internal deformation has not been found in these rocks: in the Alba formation only a very low-grade internal deformation can be recognized (stylolitic seams); on the other hand, Candas redbeds are formed of detritic quartz grains cemented by a calcitic red matrix, the pigment of which is due to haematite. This texture does not allow an easy study of internal deformation effects. Textural goniometry on calcite grains from the matrix does not show any crystallographic fabric. We are in the process of testing such a hypothesis with laboratory experiments of deformation of magnetic material (Morash *et al.* 1981) and with a palaeomagnetic study of an area where a deformation gradient has been discovered (Cogné, Bonhommet & Cobbold 1982). While it is not possible to make a definitive choice, the data are more consistent with the synfolding hypothesis (I). In any case, a magnetization time range from Early to Middle Carboniferous reconciles the various possibilities.

Before examining declinations dispersion, let us notice that the structural position varies a lot from site to site, and not only the dip, which is obvious, but also the strike which, if we assume cylindrical folds, must be parallel to the fold axes as well as the axial planes, because fold axes are horizontal. This implies local perturbations of the regional structural trend, which may be associated with the late phase of folding reported for the Asturian arc in structural studies (Julivert 1971). Moreover, the angle between the declination of the magnetic vector and the direction of the bedding [e.g. (D,S) in Table 2] is rather constant from site to site (52.4° ± 5.5°), which contrasts with the declination and strike dispersion. That means that vertical-axis rotations which have perturbed the structural attitudes, may also have modified the magnetization directions by the same angles. This suggests a new correction for a complete unfolding of the structures: 'the strike correction'. First, we chose a mean reference value for the regional structural trend: here, 230° which is the direction of the tangent to the arc at Cabo de Peñas. Then, we determine for each site the angular deviation of the mean structural position from this reference value and we apply to the

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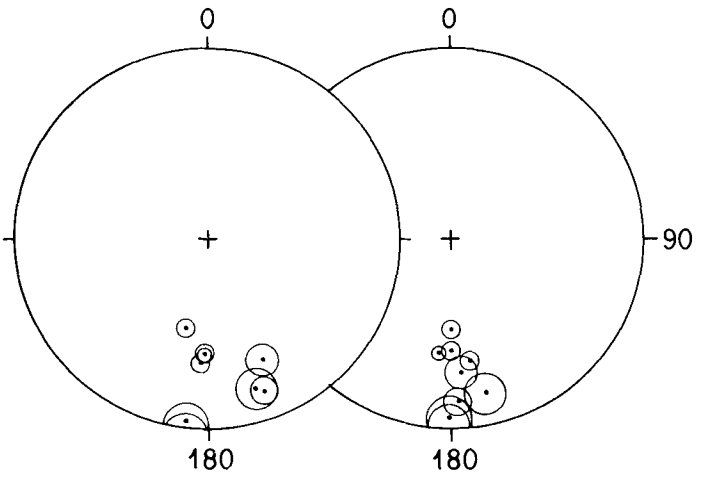


Figure 9. All Carboniferous site-mean directions after tilt correction but before (left) and after (right) 'strike' correction.

mean-site direction a vertical axis inverse rotation of this amount. To take into account the two possible interpretations developed earlier, this 'strike correction' may be applied to the 70 per cent unfolded data as well as to the 100 per cent unfolded. As the vertical axis rotations are considered to have taken place at a late stage of the deformation, this procedure should be valid in both cases. As shown in Table 2, this correction gives increases of the Fisher parameter (k) from 22 to 39 in the 100 per cent unfolding case and from 48 to 126 in the 70 per cent unfolding one. It is clear that we obtain a significant improvement in the latter case (k ratio = 2.6) but not in the former (k ratio = 1.8) at the 95 per cent confidence level (critical ratio: 2.48). One may notice, however, that a residual dispersion in the completely unfolded data (Fig. 9) still resides mainly on the inclinations which are not affected by the 'strike correction' and it is therefore not possible to get a better cluster of data by using the complete (100 per cent) tilt correction. A one-dimensional statistic on the declination may be more valuable in this case. The mean declination is $D = 175° \pm 14.1$ (1σ) before and $178° \pm 5.6$ after the strike correction. The ratio of the variance reaches 6.2, what is actually a definitive improvement. We argue therefore that we get in both cases a positive test, and this upholds the validity of the 'strike correction'.

Conclusions

This palaeomagnetic study of the Cabo de Peñas area has revealed an Ordovician magnetization. Its southerly and very steep direction is similar to previous Lower Palaeozoic data from the Armorica plate and is compatible with a position of this element near to Gondwana in Ordovician times. A Carboniferous pre- or more likely synfolding magnetization is found in sedimentary samples and also as a secondary overprint in the volcanic samples. A comparison of the site-mean directions and the structural parameters suggests that rotations around vertical axes of a later folding phase have occurred. So, a new 'strike correction' has been applied. Finally, the inclination scatter appears to correlate with the dip of the bedding. However, the repartition of our sampling sites on the two limbs of folds leads us to propose that the overall mean direction ($D = 178°$, $I = 19°$, $\alpha_{95} = 9°$) is a good approximation of the palaeomagnetic Early or Middle Carboniferous direction in Asturia. This estimation corresponds to the pre-tectonic hypothesis (2) but strictly includes the estimation

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resulting from the synfolding one (1) ($D = 176°$, $I = 19°$, $\alpha_{95} = 5°$) and was retained as no definitive proof of either interpretation has been obtained. That direction has been used in a geodynamic discussion (Perroud & Bonhommet 1981), together with other results from sites situated all along the Ibero-Amorican arc. In that paper, we proposed that the Ibero-Armorican arc is partially secondary and that the closure rotations have taken place mainly in the south branch of the arc.

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